

NEXOVA TECHNOLOGIES PVT. LTD.

System Architecture Document

NexaHR — B2B SaaS HR Management Platform

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Field	Detail
Document Version	3.2
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Audience	Internal Engineering Team & Senior Technical Stakeholders
Owner	Engineering Architecture Team, Nexova Technologies Pvt. Ltd.
Review Cycle	Quarterly — Next Review: May 2025
Legal Escalation	a.naqvi@nexova.io (Ayesha Naqvi, Legal Counsel)
Finance Approvals	b.chaudhry@nexova.io (Bilal Chaudhry, Finance Manager)

1. System Overview

NexaHR is Nexova Technologies' flagship B2B SaaS product — a cloud-native, enterprise-grade Human Resource Management System (HRMS) designed to unify and automate the full employee lifecycle for large organizations.

1.1 Core Functional Modules

- Payroll Processing — Multi-currency payroll engine with tax compliance, deductions, and disbursement automation
- Leave Management — Policy-driven leave accrual, approval workflows, and calendar integration
- Performance Management — Goal setting, 360° reviews, KPI tracking, and appraisal cycles
- Recruitment & ATS — Job posting, applicant tracking, interview scheduling, and offer management
- Onboarding — Digital document collection, task checklists, and new-hire provisioning workflows

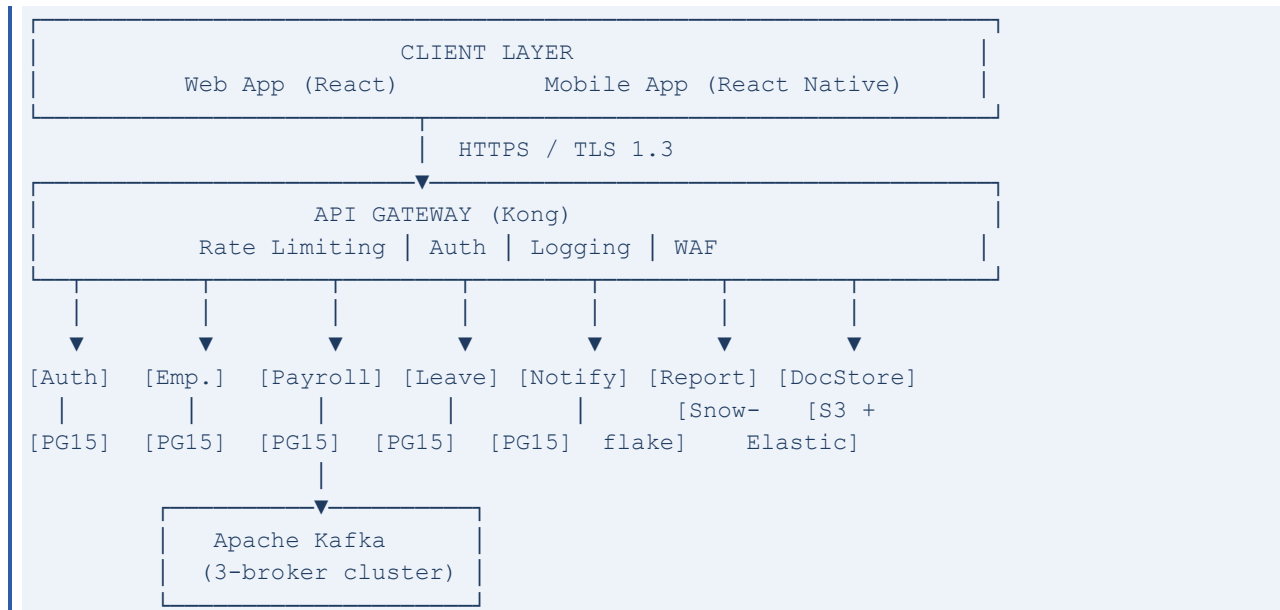
1.2 Scale & SLA

Metric	Value
Enterprise Clients	180
Total End Users	~85,000
Peak Concurrent Users	12,000 (payroll processing windows)
Availability SLA	99.95% uptime
Target RTO	< 30 minutes
Target RPO	< 5 minutes

2. High-Level Architecture

NexaHR is built on a microservices architecture, with each domain decomposed into an independently deployable service with its own dedicated data store. Services communicate asynchronously via Apache Kafka for event-driven workflows and synchronously via REST/gRPC for low-latency queries.

2.1 Architecture Diagram



2.2 Service Catalogue

Auth Service

Attribute	Detail
Responsibility	Authentication (JWT issuance, OAuth2 PKCE), session management, RBAC, SSO via SAML 2.0

Attribute	Detail
Database	PostgreSQL 15 (RDS Multi-AZ) — users, roles, sessions, tokens
Tech Stack	Node.js (TypeScript), Passport.js, Redis for token blacklisting
Team Ownership	Platform Engineering

Employee Service

Attribute	Detail
Responsibility	Master employee record management, org hierarchy, contract data, employment history
Database	PostgreSQL 15 (RDS Multi-AZ); Elasticsearch for full-text employee search
Tech Stack	Java 21 (Spring Boot 3), REST API
Team Ownership	Core HR Squad

Payroll Processing Service

Attribute	Detail
Responsibility	Payroll run orchestration, tax calculation engine, multi-currency disbursement, payslip generation
Database	PostgreSQL 15 (RDS Multi-AZ, encrypted); audit log to Elasticsearch
Tech Stack	Java 21 (Spring Boot), Apache Kafka consumer/producer
Team Ownership	Payroll Squad
⚠ Known Issue	Currently uses synchronous processing for large payrolls — see Section 8 (Technical Debt)

Leave Management Service

Attribute	Detail
Responsibility	Leave policy configuration, request/approval workflows, accrual calculations, calendar sync
Database	PostgreSQL 15
Tech Stack	Python 3.12 (FastAPI)
Team Ownership	Core HR Squad

Notification Service

Attribute	Detail
Responsibility	Multi-channel delivery: email (AWS SES), SMS (Twilio), in-app (WebSocket push)
Database	PostgreSQL 15 (delivery logs, templates); Redis for queuing
Tech Stack	Node.js (TypeScript), Bull queue
Team Ownership	Platform Engineering

Reporting & Analytics Service

Attribute	Detail
Responsibility	Pre-built HR dashboards, custom report builder, scheduled exports, workforce analytics
Database	Snowflake (data warehouse); reads from Kafka-fed pipelines
Tech Stack	Python 3.12, dbt for transformations, Apache Superset for embedded dashboards
Team Ownership	Data Engineering

Document Storage Service

Attribute	Detail
Responsibility	Secure upload, versioning, and retrieval of HR documents (contracts, payslips, IDs)
Database	AWS S3 (primary storage) with Elasticsearch metadata index
Tech Stack	Go, presigned S3 URLs, virus scanning via ClamAV
Team Ownership	Platform Engineering

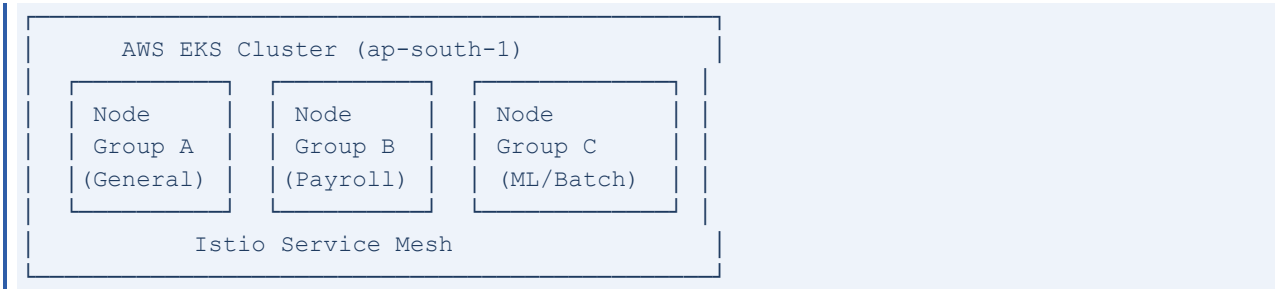
3. Infrastructure

3.1 Cloud & Regions

NexaHR operates across two AWS regions with GCP as an auxiliary ML compute environment, in Active-Active configuration with Route 53 latency-based routing and health check failover.

Provider	Region	Purpose
AWS	ap-south-1 (Mumbai)	Primary production workloads; serves APAC/MENA clients
AWS	eu-west-2 (London)	EU data residency; serves European clients (GDPR boundary) — aligned with Nexova London office
GCP	us-central1	ML model training for predictive analytics features

3.2 Containerization & Orchestration



Component	Technology	Notes
Container Runtime	Docker	Standardized base images via internal registry
Orchestration	Kubernetes (AWS EKS)	Separate clusters per region
Service Mesh	Istio	mTLS between services, circuit breaking, retry policies
CDN	AWS CloudFront	Static assets and API response caching
DNS	AWS Route 53	Health checks and automated failover

4. Data Architecture

4.1 Primary Databases

Each microservice owns its PostgreSQL 15 instance (RDS Multi-AZ), enforcing strict data isolation and preventing cross-service direct DB queries.

4.2 Caching Strategy — Redis Cluster (ElastiCache)

Data Type	TTL
User sessions	8 hours
Employee lookup results	5 minutes
Leave balance snapshots	1 minute
Static reference data (org chart)	24 hours

4.3 Search

Elasticsearch cluster (3 nodes) powers employee full-text search and stores audit logs. Indexed via Kafka consumers on data change events.

4.4 Data Warehouse

Snowflake serves as the analytics layer. Raw events stream from Kafka → S3 (data lake) → Snowflake via dbt pipelines. Reporting Service queries Snowflake exclusively — no production DB reads for analytics.

4.5 Event Streaming — Apache Kafka (3-Broker Cluster, Kraft Mode)

- Payroll events → Notification Service
- Employee record changes → Elasticsearch sync
- Leave approvals → Calendar sync, Notification Service
- Audit events → Elasticsearch, Snowflake

4.6 Backup & Recovery

- Automated daily RDS snapshots — 30-day retention
- Cross-region replication: Mumbai ↔ London
- S3 versioning enabled on all document buckets; lifecycle policy moves to Glacier after 90 days

5. Security Architecture

NexaHR enforces a Zero-Trust Network Model — no service is implicitly trusted regardless of network position.

5.1 Security Controls

Layer	Control
Network Perimeter	AWS WAF (custom rules), AWS Shield Standard

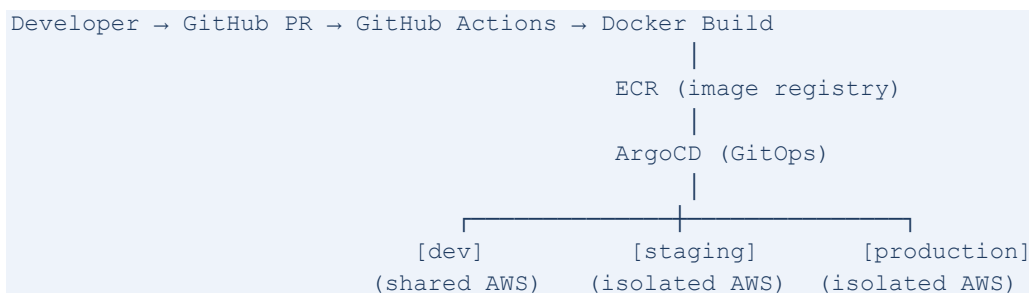
Layer	Control
API Gateway	Kong — rate limiting (per client/IP), JWT validation, request logging
Inter-Service	Istio mTLS — all service-to-service traffic encrypted
Secrets Management	AWS Secrets Manager — automatic rotation for DB credentials and API keys
Data at Rest	AES-256 on all RDS instances, S3 buckets, and EBS volumes
Data in Transit	TLS 1.3 enforced; TLS 1.2 and below rejected
Identity	JWT (15-min access tokens) + Refresh tokens (30-day, rotated)
Compliance	SOC 2 Type II (annual audit), GDPR data residency controls per region

5.2 Penetration Testing

External penetration testing is conducted quarterly by an accredited third-party vendor. Critical findings are remediated within 14 days; high-severity findings within 30 days. Results are reviewed by Ayesha Naqvi (Legal Counsel — a.naqvi@nexova.io) and Usman Baig (VP Operations).

6. CI/CD Pipeline

6.1 Pipeline Flow



6.2 Branching Strategy

GitFlow model: main (production), develop, release/*, hotfix/*, feature/* branches.

6.3 Deployment Strategies

Service	Strategy	Rationale
Auth, Employee, Leave, Notification, Document	Blue-Green	Zero-downtime, instant rollback

Service	Strategy	Rationale
Payroll Processing	Canary	Gradual traffic shift; financial data sensitivity
Reporting & Analytics	Rolling	Stateless consumers; availability-tolerant

6.4 Rollback

Automated rollback triggered on consecutive health check failures post-deployment. ArgoCD reverts to the last known-good image tag. Mean time to rollback target: < 3 minutes.

7. Observability Stack

Component	Tool	Function
Metrics	Prometheus + Grafana	Service metrics with SLI/SLO dashboards; reviewed in weekly engineering syncs
Logging	ELK Stack (Elasticsearch + Logstash + Kibana)	Centralized structured logging; PII fields masked at Logstash layer
Distributed Tracing	Jaeger	Full trace capture; 5% baseline sampling, 100% on error paths
Alerting & On-Call	PagerDuty	1-week on-call rotation per squad; escalation to Usman Baig (VP Operations)

7.1 SLI / SLO Definitions

Service	SLI	SLO
Auth Service	Login success rate	≥ 99.9% over 30 days
Payroll Service	Payroll run completion rate	≥ 99.95% over 30 days
API Gateway	P99 latency	≤ 500 ms
Document Storage	Upload success rate	≥ 99.5%
Notification Service	Email delivery rate	≥ 98% within 60 seconds

8. Known Technical Debt & Roadmap Items

The following items are tracked in the engineering backlog. Severity ratings: ● High | ● Medium-High | ● Medium.

ID	Issue	Severity	Resolution
TD-01	Payroll Service: synchronous processing for large payrolls (>500 employees). Causes API timeout risk during peak payroll windows and blocks concurrent runs. Migration to async Kafka-based pipeline in progress.	● High	Q3 2025
TD-02	Employee Service: N+1 query issue on org-chart deep-fetch endpoint. Under load with deep hierarchies (>8 levels), response times degrade significantly. Requires recursive CTE optimization.	○ Medium	Q2 2025
TD-03	Auth Service: SAML 2.0 library (passport-saml) is 3 major versions behind. Security patching blocked by breaking API changes. Upgrade requires SSO re-testing with affected enterprise clients.	● High	Q2 2025
TD-04	Notification Service: no dead-letter queue (DLQ) for failed SMS events. Failed SMS notifications are silently dropped rather than retried or logged for manual resolution.	○ Medium	Q2 2025
TD-05	Reporting Service: Snowflake transformation pipelines lack incremental load logic. Full table scans on large fact tables cause daily dbt runs to exceed the 2-hour window at month-end peaks.	● Medium-High	Q3 2025

⚠ Priority Action — TD-01 & TD-03

TD-01 (Payroll async migration) and TD-03 (SAML library upgrade) are both HIGH severity. Engineering leads must confirm Q2/Q3 2025 delivery commitments in the next sprint planning session. Any delays must be escalated to Usman Baig (VP Operations) and reviewed by Ayesha Naqvi (Legal Counsel) for any compliance implications.